

Node.js and MongoDB Notes

Node.js is an open-source JavaScript runtime built on Chrome's V8 engine. It allows developers to build fast and scalable server-side applications. Node.js uses an event-driven, non-blocking I/O model, making it suitable for APIs, real-time applications, chat applications, and microservices.

MongoDB

MongoDB is a NoSQL document database that stores data in BSON documents instead of tables. It is schema-flexible, horizontally scalable, and designed for high-performance applications. Common use cases include content management systems, e-commerce, analytics, and IoT platforms.

MongoDB Hosting

MongoDB hosting means running a MongoDB database on a server so applications can connect to it. Hosting options include: 1. Local installation on a developer machine. 2. Self-hosted on a VPS or cloud VM. 3. Managed cloud hosting using MongoDB Atlas.

Why use MongoDB Atlas?

MongoDB Atlas is the official cloud-hosted MongoDB service. Benefits include automatic backups, monitoring, security, scaling, high availability, replica sets, automatic updates, and global deployment.

Docker

Docker packages applications into lightweight containers. Containers ensure that applications run consistently across development, testing, and production environments.

REST API

A REST API exposes resources over HTTP using methods such as GET, POST, PUT, and DELETE. Spring Boot and Node.js are both commonly used to build REST APIs.

Vector Databases

Vector databases store embeddings produced by AI embedding models. Popular vector databases include Qdrant, Pinecone, Weaviate, Chroma, and Milvus. They are widely used in Retrieval-Augmented Generation (RAG) systems for semantic search.

RAG Workflow

Load documents → Split into chunks → Generate embeddings → Store in Qdrant → Convert user query into embedding → Retrieve similar chunks → Send retrieved context to an LLM → Generate the final answer.